

Md Abu Sayeam Mondol Shejan

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PROFESSIONAL SUMMARY

AI Engineer experienced in designing, developing, and deploying production AI applications across generative AI, large language models (LLMs), retrieval-augmented generation (RAG), multi-agent systems, natural language processing, and computer vision. Strong Python background with hands-on experience in QLoRA/PEFT fine-tuning, prompt engineering, embeddings, semantic search, vector databases, model evaluation, FastAPI services, Docker, cloud deployment, and CI/CD. Builds modular, maintainable AI systems and collaborates with cross-functional teams to translate business requirements into reliable products for international clients.

EDUCATION

North South University	2026
Bachelor of Science in Computer Science and Engineering	Dhaka, Bangladesh
Cantonment Public School and College, Lalmonirhat	2020, 2018
Higher Secondary Certificate and Secondary School Certificate, Science	

PROFESSIONAL EXPERIENCE

AI Engineer	Sep 2026 – Present
<i>Omicon Group</i>	<i>Dhaka, Bangladesh / Full-time, On-site</i>

- Engineer enterprise-grade, locally fine-tuned LLM solutions with advanced tool-calling capabilities, optimized for Bengali, Banglish, and English understanding and multilingual embedding-based retrieval.
- Build evaluation datasets and testing frameworks to benchmark LLM, RAG, retrieval, and tool-calling performance; continuously optimize models and pipelines for accuracy, reliability, and cost-efficient inference.
- Contribute to AI-powered catalogue automation for **Boibazar.com** and develop an agentic customer-facing chatbot for intelligent product discovery, assistance, and workflow automation.
- Lead research, development planning, and system architecture design for **Lecture Publications' e-learning platform**, defining AI capabilities, core components, data flows, integrations, and implementation roadmaps.
- Manage and scale local AI/GPU infrastructure for high-volume production systems, monitoring model performance, resource utilization, latency, and reliability while optimizing deployment and inference workloads.

AI Engineer	Jan 2026 – Aug 2026
<i>Softvence Agency</i>	<i>Dhaka, Bangladesh / Full-time, On-site</i>

- Led AI project architecture and implementation planning in collaboration with UI/UX, frontend, and backend teams, translating product requirements into system components, integrations, and delivery plans.
- Worked directly with international clients to understand business and technical requirements, define feasible AI solutions, clarify trade-offs, and plan implementation milestones.
- Coordinated cross-functional execution, tracked delivery progress, resolved technical dependencies, and provided clients with clear project updates throughout the development lifecycle.
- Guided the delivery of agentic AI applications, RAG systems, automation pipelines, scalable APIs, and production deployments focused on reliability and business impact.

Jr. AI Engineer	Jul 2025 – Dec 2025
<i>Softvence Agency</i>	<i>Dhaka, Bangladesh / Full-time, On-site</i>

- Developed production AI solutions across machine learning, deep learning, large language models, and computer vision using modern frameworks and tools.
- Built agentic AI applications, RAG systems, automation pipelines, and scalable APIs with LangChain, n8n, FastAPI, Docker, and vector databases.
- Deployed production-ready AI systems focused on scalability, cost efficiency, reliability, and measurable business impact.

TECHNICAL SKILLS

Programming and Software Engineering: Python, C, C++, Java, JavaScript, TypeScript, Bash, Git, API design, testing, documentation, modular architecture

Generative AI and Agents: LLMs, prompt engineering, prompt optimization, RAG, AI assistants, chatbots, tool calling, multi-agent systems, LangChain, LangGraph, Ollama, Llama, Qwen

Fine-Tuning and NLP: Hugging Face Transformers, TRL, PEFT, LoRA, QLoRA, supervised fine-tuning, 4-bit quantization, GGUF, transformer architecture, attention, NLP

Retrieval and Vector Databases: Embeddings, semantic search, knowledge retrieval, ChromaDB, FAISS, Pinecone

Machine Learning and Vision: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, NumPy, Pandas, deep learning, computer vision, model evaluation

Backend and Data: FastAPI, Flask, Django, REST APIs, Swagger, PostgreSQL, MySQL, MongoDB

MLOps, Cloud and Deployment: MLflow, Docker, AWS EC2, Google Cloud, GitHub Actions, Bitbucket, CI/CD, Nginx, Apache, VPS deployment

Automation and Tools: n8n, Zapier, GitHub, Postman, PowerShell

SELECTED PROJECTS

APEX Hybrid AI Lab: Local Multi-Agent Orchestration Framework GitHub

- Built a secure, local-first autonomous agent framework with an OpenClaw master router that performs tool calling and delegates requests to specialized agents powered by local Ollama LLMs.
- Implemented structured JSON ReAct execution, critic-based self-reflection and recovery, response validation, ChromaDB conversational memory, context compression, and scheduled workflows.
- Integrated PC control, Playwright browser automation, a bidirectional Telegram assistant, proactive notifications, and Docker deployment through modular Python components.
- **Technologies:** Python, Ollama, ChromaDB, Playwright, APScheduler, Telegram Bot API, Docker

Qwen3-14B Fine-Tuning for Reasoning and Conversational AI GitHub

- Built an end-to-end supervised instruction-tuning and evaluation pipeline for Qwen3-14B using QLoRA and Unsloth for reasoning and conversational tasks.
- Reduced training memory through 4-bit quantization, gradient checkpointing, and LoRA adapters with Hugging Face Transformers, TRL, and PEFT.
- Exported the optimized model to GGUF for local inference and deployment with llama.cpp and Ollama.
- **Technologies:** Python, Qwen3-14B, Unsloth, Transformers, TRL, PEFT, LoRA, Ollama

PhyChat: Local Retrieval-Augmented Generation Chatbot GitHub

- Built a local intelligent document processing and PDF question-answering application using Llama 3.2, LangChain, Ollama embeddings, and ChromaDB.
- Implemented document ingestion, chunking, embedding generation, semantic search, context retrieval, streaming responses, and an interactive Streamlit interface.
- **Technologies:** Python, LangChain, Llama 3.2, ChromaDB, Ollama, Streamlit

Brain Tumor Segmentation from MRI with Knowledge Distillation GitHub

- Developed and evaluated deep learning MRI tumor segmentation models with U-Net, EfficientNetB7, and ResUNet using the BraTS dataset.
- Applied preprocessing, augmentation, and knowledge distillation; measured model quality with Dice score and intersection over union (IoU).
- **Technologies:** Python, TensorFlow, Keras, OpenCV, NumPy, Matplotlib

AWARDS AND COMPETITIONS

Infinity AI BuildFest, Cloud Camp BD – Finalist Jun 2026

Built and presented EduOrbit, a full-stack AI education platform featuring gamified learning, personalized study assistance, and career-readiness support. [Event Link](#)

WebXtream Hackathon, North South University – Finalist Jul 2025

Developed and presented a working web technology solution through rapid iteration and cross-functional collaboration under competition deadlines. [Project Link](#)

National RoboFest, East West University – Finalist Sep 2024

Collaborated on robotics problem-solving and hardware–software integration in a national-level competition. [Event Link](#)